

time-of-flight (ToF) lidar, which performs less optimally under certain conditions, more prevalent.

The choice of wavelength—1550 nanometres versus 905 nanometres—also presents dilemmas. While 1550 nanometres is eye-safe and supports longer distances due to higher power emissions, it requires more complex, costly technologies. Conversely, 905 nanometres, though more economical, raises safety issues for eye exposure and typically has shorter ranges.

Furthermore, lidar's broader adoption is stalled by maturity and integration challenges. Many applications need further advances to ensure lidar systems can integrate seamlessly with existing technologies while remaining reliable and efficient.

Competition from other sensors

Cameras and radar have long been staples in automotive sensing, prized for their cost-effectiveness and robust performance under diverse conditions. Cameras excel in visual recognition at a lower cost than lidar, capturing detailed visual data. However, their performance declines in harsh lighting and lack depth information. Radar, meanwhile, is valued for its ability to operate in adverse weather and detect distant objects, although it lacks the resolution to differentiate objects with precision.

Lidar, known for its superior depth perception and spatial data, often plays a secondary role in automotive applications. As camera and radar technologies advance in accuracy, reliability, and recognition capabilities, manufacturers prefer these more established, cost-effective solutions, relegating lidar to a backup function.

Rapid advancements in AI enable cameras to handle most scenarios, with industry leaders

Lidar market dynamics in China

- **Aggressive pricing for market leadership.** Chinese firms aggressively cut lidar prices, often selling at a loss to dominate the market, scale production, and penetrate the mass market.
- **Advancements and cost efficiency.** Improved manufacturing processes and technological advancements in lidar reduce costs and enhance its appeal for cost-conscious automakers.
- **Premium branding for safety.** Lidar-equipped vehicles are marketed as premium, enhancing perceptions of safety and technology to justify higher prices and establish market leadership.
- **Crucial for higher autonomy.** Lidar is essential for higher levels of vehicle autonomy (Levels 4 and 5), supporting safe operation without human input and paving the way for full automation.
- **Sensor integration enhances safety.** Lidar complements other sensors like cameras and radar, creating a comprehensive detection system that improves safety and decision-making.
- **End-user focus on safety.** Integrating various sensors aims to enhance safety, which is the top priority for consumers, ensuring robust performance and preventing accidents.

such as Tesla promoting AI-powered cameras. However, relying solely on cameras can pose risks, particularly in low visibility or rapidly changing conditions. To address this, manufacturers enhance system reliability through sensor fusion—combining multiple sensor types, including the more expensive lidar—to ensure safety redundancy. They must balance cost, performance, and safety when selecting sensor technologies.

Lidar in autonomous driving

Lidar's role in autonomous driving remains a point of contention, balancing its proven benefits against evolving technologies. Known for reliable depth data and enhancing safety, lidar is considered in many scenarios. Yet, it struggles in extreme weather, as rain, snow, and fog can reduce its accuracy. This limitation is critical for vehicles in adverse climates, suggesting that complementary sensors are needed.

Furthermore, lidar systems vary widely. While long-range sensors detect objects up to 200 metres away, suitable for highways, other types, like flash lidar, are limited to shorter ranges, affecting their utility in different driving conditions.

Concurrently, advances in AI are boosting camera and radar capabilities, potentially decreasing lidar dependency. This is particularly relevant for Level 4 and Level 5

autonomous vehicles, where lidar is a safety net against sensor failures.

Many companies seek a mixed-sensor strategy, combining various sensors to optimise performance, reliability, and cost. This approach varies by vehicle type, from comprehensive coverage in robotaxis to front-facing sensors in consumer vehicles for collision avoidance.

Despite its advanced depth perception and precision mapping capabilities, lidar technology faces significant hurdles in widespread adoption within the automotive industry due to high costs, technical challenges, an immature supply chain, and complex integration processes. While lidar remains superior in highly-reliability scenarios, the presence of cost-effective alternatives like cameras and radar further complicates its mainstream integration. As the technology evolves and potentially becomes more cost-effective, lidar may see increased use in high-end vehicles, autonomous taxis, and other sectors such as industrial automation and robotics, where its detailed spatial mapping can be fully utilised. The balance of performance, cost, and practical integration will be crucial for lidar's future in automotive applications and beyond. **EFY**

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